Cisco PE Optical Transport Failure SOP (ALARM_9067)

Node: KR-PE-15 | **Vendor:** Cisco | **Severity:** Critical | **Alarm Code:** ALARM_9067

Description

The Optical Transport Failure on the Cisco PE (KR-PE-15) at KR Puram Access is a critical alarm that occurs due to degradation of SFP/XFP modules, fiber cuts, or Rx/Tx power loss. This failure domain is specific to the PE role due to its responsibility for aggregating and forwarding traffic from multiple access nodes. The Cisco PE utilizes the Optical Transport Network (OTN) protocol to transparently transport client signals over optical fibers. The OTN protocol uses SFP/XFP modules to transmit and receive optical signals. When an SFP/XFP module fails or experiences degradation, it can cause a significant impact on the overall network performance. Additionally, fiber cuts or Rx/Tx power loss can also cause the OTN protocol to fail, resulting in the critical ALARM_9067. It is essential to identify the root cause of the failure and perform the necessary corrective actions to prevent further network disruptions.

Pre-requisite Checks

- Check: Verify the site location and node ID (KR Puram Access and KR-PE-15, respectively) to ensure the correct equipment is being addressed., Description: This step ensures that the correct equipment is being targeted for troubleshooting and repair.

- Check: Confirm the alarm code (ALARM_9067) and severity (Critical) to determine the urgency of the issue., Description: This step verifies the severity of the alarm to prioritize the troubleshooting and repair efforts.

- Check: Perform a redundant link status check to identify if the failure is isolated to a single link or a broader network issue., Description: This step helps to determine if the failure is isolated to a single link or a more widespread network issue, which can impact the troubleshooting and repair strategy.

Potential Root Causes

Cause	Probability
SFP/XFP Module Failure	High
Fiber Cut	Medium
Rx/Tx Power Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show controllers t3/0/0 | include 'SFP Status', Expected Output: SFP Status: Not present, Description: This command checks the SFP/XFP module status on the specified interface. The expected output indicates that the SFP/XFP module is not present, which could be the root cause of the Optical Transport Failure.
```

```
Command: show controllers t3/0/0 | include 'Fiber Status', Expected Output: Fiber Status: Not present, Description: This command checks the fiber status on the specified interface. The expected output indicates that the fiber is not present, which could be the root cause of the Optical Transport Failure.

Command: show interfaces t3/0/0 | include 'Rx/Tx Power', Expected Output: Rx/Tx Power: -10dbm, Description: This command checks the Rx/Tx power levels on the specified interface. The expected output indicates that the Rx/Tx power levels are within the normal range, which suggests that the issue may not be related to power loss.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace /controllers t3/0/0 {sfp=0}, Description: This step involves replacing the SFP/XFP module on the specified interface. The command replaces the SFP/XFP module with a new one, which should resolve the Optical Transport Failure.

Step 2: Step: Verify the fiber connections, Command: show controllers t3/0/0 | include 'Fiber Status', Description: This step involves verifying the fiber connections on the specified interface. The command checks the fiber status, which should indicate that the fiber is present and connected properly.

Step 3: Step: Adjust the Rx/Tx power levels, Command: set interfaces t3/0/0 {rx-power=-5dbm, tx-power=-5dbm}, Description: This step involves adjusting the Rx/Tx power levels on the specified interface. The command sets the Rx/Tx power levels to the recommended values, which should resolve the Optical Transport Failure.

Post-Check Verification

```
Command: show controllers t3/0/0 | include 'SFP Status', Expected Output: SFP Status: Present, Description: This command checks the SFP/XFP module status on the specified interface. The expected output indicates that the SFP/XFP module is present, which verifies that the resolution step was successful.

Command: show controllers t3/0/0 | include 'Fiber Status', Expected Output: Fiber Status: Present, Description: This command checks the fiber status on the specified interface. The expected output indicates that the fiber is present, which verifies that the resolution step was successful.

Command: show interfaces t3/0/0 | include 'Rx/Tx Power', Expected Output: Rx/Tx Power: -5dbm, Description: This command checks the Rx/Tx power levels on the specified interface. The expected output indicates that the Rx/Tx power levels are within the normal range, which verifies that the resolution step was successful.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: restore /controllers t3/0/0 {sfp=0}, Description: This step involves restoring the original SFP/XFP module on the specified interface. The command restores the original SFP/XFP module, which should revert the changes made during the resolution step.

- Step: Revert the fiber connections, Command: show controllers t3/0/0 | include 'Fiber Status', Description: This step involves reverting the fiber connections on the specified interface. The command checks the fiber status, which should indicate that the fiber is disconnected properly.

- Step: Revert the Rx/Tx power levels, Command: unset interfaces t3/0/0 {rx-power, tx-power}, Description: This step involves reverting the Rx/Tx power levels on the specified interface. The command unsets the Rx/Tx power levels, which should revert the changes made during the resolution step.